PRIYA VENKATESH

Software Engineering & Cloud Systems Intern

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SUMMARY

Final-year Computer Science student with practical production experience building microservices and cloud automation pipelines. Proven ability to reduce query latency and automate CI/CD deployments.

EDUCATION

Indian Institute of Information Technology **2021 – 2025**

B.Tech in Information Technology in Information Technology *GPA / Score: 9.15 / 10.0 CGPA*

Delhi Public School **2019 – 2021**

Higher Secondary Certificate (CBSE Class XII) in Physics, Chemistry, Mathematics *GPA / Score: 96.4%*

SKILLS & COMPETENCIES

Technical Skills: Java, Spring Boot, Python, FastAPI, React, TypeScript, Kubernetes, Redis, PostgreSQL, Docker

Core Competencies: Agile / Scrum, Cross-Functional Collaboration, System Architecture

KEY PROJECTS

Distributed Distributed Task Queue Lead Architect

Engineered a fault-tolerant asynchronous job queue with dead-letter exchanges and exponential backoff retry policies.

- Benchmarked task processing throughput exceeding 12,000 tasks/second with sub-5ms Redis latency.
- Designed graceful worker shutdown ensuring zero in-flight task loss during rolling cluster updates.
- Built Prometheus metrics exporter monitoring queue lag, worker memory, and failure rates.

Tools: Go, Redis, PostgreSQL, Docker

Live Demo · **Source Code**

Real-Time Collaborative Code Editor Frontend & WebSocket Developer

Developed an in-browser multiplayer code editor with operational transformation and multi-cursor synchronization.

- Implemented Yjs CRDT protocol resolving concurrent keystrokes across geographically distributed clients.
- Added syntax highlighting for 14 programming languages with Monaco Editor integration.
- Containerized code execution sandbox executing Python, C++, and JavaScript with strict time limits.

Tools: TypeScript, React, Node.js, WebSocket, CRDT

Live Demo · **Source Code**

Smart Campus Shuttle Tracker Mobile & Backend Developer

GPS telemetry platform tracking 12 campus transit shuttles in real time for 8,500 daily university commuters.

- Streamed vehicle coordinates over MQTT with sub-second websocket client broadcast.
- Reduced student wait times by an estimated 18 minutes through predictive arrival algorithms.

Tools: FastAPI, PostgreSQL, Leaflet.js, MQTT

Source Code

PROFESSIONAL EXPERIENCE

Backend Software Engineering Intern, Razorpay Technologies **May 2024 – Jul 2024**

Bangalore, India

Worked with the Merchant Settlements team optimizing payment reconciliation batches and idempotency safeguards.

- Refactored batch settlement SQL queries reducing nightly execution window from 42 minutes to 16 minutes.
- Implemented Redis distributed locks preventing duplicate payout dispatches under concurrent webhooks.
- Wrote 45+ unit and integration tests achieving 91% code coverage on settlement reconciliation modules.

Student Research Assistant, Cloud Computing Research Lab **Jan 2024 – May 2024**

IIIT Hyderabad

Conducted research on serverless function cold-start latency mitigation under fluctuating traffic patterns.

- Benchmarked OpenFaaS container provisioning times across varied Linux kernel cgroup configurations.
- Co-authored research paper draft accepted at National Student Cloud Computing Symposium.

CERTIFICATIONS

Kubernetes and Cloud Native Associate (KCNA) — Linux Foundation & CNCF 2024

Oracle Certified Professional: Java SE 11 Developer — Oracle 2023